MILITARY SPECIFICATION

TRANSFORMER, IMPEDANCE MATCHING (BALUN), RECEIVE ONLY

1. SCOPE

1.1 Balanced to unbalanced (balun) transformers covered by this specification are to be utilized in a receiving system providing an efficient impedance match and transfer of received signals from a balanced antenna to an unbalanced transmission line without introducing undesirable intermodulation products.

1.2 Classification. Balun transformers covered by this specification shall be of the following types.

Type I. - 50 ohms primary impedance unbalanced; 50 ohms secondary impedance balanced.
Type II. - 50 ohms primary impedance unbalanced; 200 ohms secondary impedance balanced.
Type III. - 50 ohms primary impedance unbalanced; 300 ohms secondary impedance balanced.
Type IV. - 50 ohms primary impedance unbalanced; 600 ohms secondary impedance balanced.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of
MIL-I-45208 - Inspection System Requirements

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-810 - Environmental Test Methods

HANDBOOKS

MILITARY

MIL-HDBK-216 - RF Transmission Lines and Fittings

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The balun shall be in accordance with MIL-E-16400 to the extent specified and the following requirements and be able to perform under any combination of service conditions specified.

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3.1.1 Description. The balun offered shall be designed for broad band operation. It shall be suitable for converting the characteristic impedance of high frequency antennas without adjustment. The balun furnished under this specification shall be complete and ready to install.

3.1.2 Definitions. The definition of terms used in this specification are intended to be in accordance with MIL-E-16400.

3.1.3 Life. The equipment shall be designed to render trouble-free performance with little or no maintenance for a period of at least 15 years.

3.1.4 Environmental conditions. The balun shall be capable of continuous reliable operation under temperature, humidity, wind speed and icing conditions for class 1 equipment as specified in MIL-E-16400 and solar radiation conditions encountered in the earth's atmosphere. When subjected to the nonoperating conditions for temperature, the equipment shall not be damaged nor its performance degraded.

3.1.5 Mechanical design. The balun shall conform to the mechanical design requirements of MIL-E-16400 regarding construction, welding and soldering. The balun shall conform to MIL-STD-810 for ground equipment and for military application regarding shock, vibration, and altitude (see 4.5.11).

3.1.5.1 Drilling. Holes shall be cleanly drilled and burrs removed.

3.1.5.2 Riveting. Riveting shall not be used for mounting parts.

3.1.5.3 Securing of parts. Brackets, lugs, flanges, inserts, bolts, and other mounting hardware or accessories shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the items in which they are used are subjected to specified service conditions. Friction between mating surfaces shall not be employed as the sole means of preventing movement of fixed parts. Parts shall be secured in such manner that failure of a single screw will not free the part completely.

3.1.5.4 Brazing. Brazing of steel, copper alloys and nickel alloys shall conform to Type V, Grade B in accordance with MIL-B-7883.

3.1.5.5 Enclosure. The degree of enclosure shall be spraytight as specified in MIL-STD-108.

3.1.6 Electrical design. The electrical design of the balun shall be in accordance with applicable portions of MIL-E-16400 unless otherwise specified herein.

3.1.6.1 Wiring and cabling. Internal wiring and cabling shall be neat, sturdy and secured so that intermittent contact, vibration and internal part movements shall not occur and impair performance.

3.1.6.1.1 Protection. Wires and cables shall be so placed and protected to avoid rough or irregular surfaces or sharp edges. Where wires and cables are held in place by metal partitions, shields, insulators, or similar items, they shall be protected by grommets, or bushings satisfactory to the command or agency concerned.

3.1.6.2 Grounding. An external grounding terminal shall be provided on the balun. The ground terminal shall be at least a 3/8-inch diameter threaded stud plus attaching hardware in accordance with MIL-E-16400.

3.1.6.3 Insulators. The insulators to be employed on the balanced side shall be spaced not to exceed 12 inches and shall not be less than 2 inches apart relative to their geometric centers.

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3.1.7 Parts. Electronic parts and mechanical hardware selection and use, shall conform to the requirements of MIL-E-16400 unless specifically stated otherwise herein. The use of standard parts as defined in MIL-E-16400 does not relieve the contractor of the responsibility to conform to the stated performance requirements of this specification.

3.1.7.1 Parts interchangeability. Parts interchangeability shall be in accordance with requirement 7 of MIL-STD-454.

3.1.8 Materials. Materials shall conform to the requirements of MIL-E-16400 unless otherwise stated herein.

3.1.8.1 Unspecified definite material. When a definite material is not specified, a material shall be used which will enable the balun transformer to meet the requirements specified herein. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product. In addition, materials which are capable of producing dangerous gasses or other harmful toxic effects under conditions, including fire, encountered in Naval Service shall not be used, unless suitable non-toxic material is not available. Toxic materials shall not be incorporated in the design without prior approval of the command or agency concerned.

3.1.8.1.1 Fiberglass. Fiberglass shall not be used in an external environment where exposure to sun light and weathering would otherwise degrade its electrical properties.

3.1.8.2 Freedom from defects. All materials shall be free from harmful seams, laminations, cracks, dents, kinks and other injurious defects.

3.1.8.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined MIL-E-16400 shall not be used in intimate contact with each other. Where it is necessary that a combination of dissimilar metals be assembled together, an interposing metal compatible with each shall be used.

3.1.8.4 Aluminum alloys. Parts made of aluminum and aluminum alloys shall be suitably protected in accordance with MIL-E-16400.

3.1.8.5 Brittle materials. Brittle castings, or parts made of ceramic or other brittle material, shall not be over-constrained by the means used to secure them. Mating surfaces shall be accurately machined to ensure 360 degree contact between the brittle material and the other surface. Mounting washers of suitable plastic with long-life characteristics or soft copper having slight compressibility shall be provided to prevent breakage or cracking of the parts. Lead washers shall not be used.

3.1.9 Gaskets. Gaskets shall be in accordance with the requirements of MIL-E-16400.

3.1.10 Coatings and treatments. Coatings and treatments shall be in accordance with MIL-E-16400. In addition, exposed parts, except insulators and coaxial connections, shall be primed and painted light-gray enamel in accordance with MIL-E-16400. Polyurethane paints may be considered for this application provided written approval is first obtained from the cognizant Government agency. Request for such approval must contain the following information as a minimum; environmental data, tests performed and results.

3.1.11 Supplementary requirements. Supplementary requirements of MIL-E-16400 regarding serial numbers, date of manufacture, instruction plates, Government nomenclature, and identification and markings shall be complied with unless otherwise specified (see 6.2.1). A schematic diagram of the balun circuit shall be marked in plain view on the cover of the unit.

3.1.12 Thermal design. Thermal design shall be in accordance with MIL-E-16400.

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3.2 Detailed requirements.

3.2.1 Weight and dimensions. The overall weight and dimensions of the balun shall be an absolute minimum commensurate with good engineering and the requirements of this specification. Maximum weight shall be 20 pounds. Maximum dimensions shall be a height of 20 inches and a diameter, width or depth of 12 inches.

3.2.1.1 Mounting. The balun transformer shall be furnished with means for wooden pole or metal tower mounting.

3.2.2 Electrical.

3.2.2.1 Frequency range. Frequency range of the balun shall be 2 to 32 megahertz (MHz).

3.2.2.2 Matching. The balun shall be capable of matching a balanced secondary load to an unbalanced primary load (see 1.2) over the frequency range 2 to 32 MHz.

3.2.2.3 Insertion VSWR. The insertion VSWR shall not exceed 1.25:1 over the operating frequency range.

3.2.2.4 Unbalance. The degree of unbalance shall not exceed 5 percent over the frequency range of the balun, based on RF current or voltage readings in each line of the balanced pair.

3.2.2.5 Connectors. The connectors (on the coaxial side) shall be LC type in accordance with MIL-HDBK-216 mounted on the bottom surface of the vertically mounted balun.

3.2.2.6 Lightning protection. The balun transformer shall include lightning protection at the high impedance secondary in the form of maintainable spark gaps to case ground.

3.2.2.7 Shielding. The balun shall be shielded from stray electromagnetic fields.

3.2.2.8 Intermodulation products. The second and third order intermodulation products present at an output of the type I multicoupler when two 0.25 volt rms signals are applied to its input, shall be at least 60 dB below either of the input signals (a maximum of 250 microvolts). The second and third order intermodulation products present at outputs of the types II and III multicouplers when two 0.50 volt rms signals are applied to their inputs, shall be at least 70 dB below either of the input signals (a maximum of 158 microvolts). The two signals supplied to the input of the multicoupler shall have frequencies f_1 and f_2 in the frequency range of the multicoupler in such a relationship that the second order frequency equal to $f_1 \pm f_2$ and the third order product frequency equal to $2f_1 \pm f_2$ falls within the specified range of the multicoupler and outside its range from 0.1 MHz to 60 MHz.

3.3 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

3.4 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's inspection system. The contractor shall provide and maintain an effective inspection system acceptable to the Government covering the supplies under the contract. The inspection system shall be in accordance with MIL-I-45208. A detailed inspection system plan, as it would apply to the requirements of this specification shall be submitted to the command or agency concerned for evaluation of acceptability, prior to first article inspection (see 6.2.2). The inspection system plan shall include flow charts showing the complete flow of material from source inspection or inspection on receipt, through all manufacturing processes and testing, to final shipment. The various processes and inspection positions shall be clearly identified on the flow charts and cross referenced to the detailed inspection procedures and criteria of the planned quality control system. Any changes to the acceptable inspection system shall be submitted to the command or agency concerned for written approval prior to use.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, (c) Government testing of a first article and, (d) Government product inspection of delivered products to assure compliance with this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Test procedures. The contractor shall prepare and submit written test procedures, including test equipment and tolerance limits to be used for first article inspection, quality conformance inspection, and other testing to the command or agency concerned via the Government inspector, for review and approval prior to beginning tests (see 6.2.2). Each test shall be identified in accordance with 4.2 as to classification.

4.1.3.1 Procedures for inspections, tests and demonstrations shall contain step-by-step information which will enable the operator to properly perform to the requirements of this specification. The methods of examination shall be clearly defined.

4.1.3.2 The procedures shall contain a list of test equipment identified by name, model, and accuracy. Special test equipment shall also be listed and identified.

4.1.3.3 Data sheets. Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever quantitative measurements are made, the results shall be shown by quantitative data.

4.2 Classification of inspection. The method of examination and testing of the equipment fall within the following classifications:

(a) First article inspection
(b) Quality conformance inspection

(1) Production inspection
(2) Production control inspection
(3) Inspection of preparation of delivery

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4.3 First article inspection. Unless otherwise specified (see 6.2.1); 1 unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I.

Table I. First article inspection.

Requirement paragraph	Reference paragraph
Surface examination	4.5.1
Operating test	4.5.2
Weight and dimensions	4.5.3
Wind Speed and Icing	4.5.5
Enclosure test	4.5.6
Temperature test	4.5.7
Humidity test	4.5.8
Shock, vibration and altitude	4.5.11
Salt spray test	4.5.12
Thermal design	4.5.13
Solar radiation test	4.5.14

4.3.1 First article test report. A first article test report shall be prepared and submitted to the Government covering the results of the contractor's first article inspection (see 6.2.2).

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4.1.3 and shall include the requirements of table II.

Table II. Production inspection.

Requirement	Reference
Surface examination	4.5.1
Operating test	4.5.2

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall be approved in accordance with 4.1.3. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

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Table III. Production control inspection.

Inspection	Inspection paragraph	AQL Percent
Weight and dimensions	4.5.3	6.5
Enclosure test	4.5.6	6.5

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Nonconformance and retest. If a sample unit fails a test specified in 4.4.2 the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, and processes; the equipment shall then be retested.

4.4.4 Corrective action. If corrective action results in a change of part(s), electrical/mechanical layout or function, or if the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred immediately to the contracting officer. It shall then be determined by the command or agency concerned whether correction should be made to the equipments already delivered under the contract.

4.4.5 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to and have passed production control inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.4.6 Quality conformance inspection report. The contractor shall submit to the command or agency concerned a complete report detailing the results of all testing performed (see 6.2.2). The report shall list the serial numbers of equipments tested during the reporting period. During the conduct of tests, performance parameters shall be measured and verified in accordance with this specification. Equipment will not be accepted unless the report is complete.

4.5 Inspection procedures.

4.5.1 Surface examination. Equipment shall be examined for conformance to 3.1.5 through 3.1.8, 3.1.10, 3.1.12, and 3.3 for the following:

(a) Workmanship, assembly, and fit, mechanical safety, and marking
(b) Materials, parts, and finish
(c) Treatment for prevention of corrosion

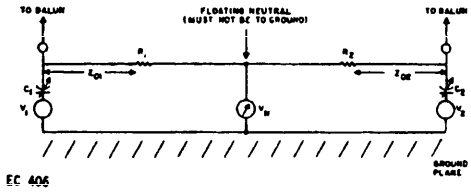
4.5.2 Operating test. The equipment shall be energized and subjected to an operating test to ensure qualitatively conformance to safety requirements and the proper functioning of the equipment, including as a minimum, the following:

4.5.2.1 Matching. Testing shall be conducted at 2, 4, 6, 8, 12, 15, 18, 28, 30, and 32 MHz for first article inspection and at 2, 4, 9, 18, and 32 MHz for quality conformance inspection to demonstrate compliance with 3.2.2.2.

4.5.2.2 Unbalance. The degree of unbalance shall be measured over the frequencies specified in 4.5.2.1 for conformance to 3.2.2.4. The degree of unbalance shall be measured as follows: The termination assembly (see figure 1) consists of selected resistors R_1 and R_2 which are connected to the input terminals via transmission lines Z_{o1} and Z_{o2}. The characteristic impedance of Z_{o1} and Z_{o2} is established by the diameter of

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the connecting wire used and its proximity to the ground plane. The junction of R_1 and R_2 is the neutral connection. Capacitors C_1 and C_2, approximately 0.6 pf provide isolation and ensure that impedance loading effects of V_1 and V_2 are minimal. The division ratio must be established in order to relate voltmeter readings to actual terminal to ground voltage.



V_1, V_2 G.R. model 1800A VTVM or equivalent

V_N Boonton model 91CA VTVM or equivalent

C_1, C_2 JFD Piston Trimmer 0.6 pf approx.

$R_1 \cong R_2 \cong Z_{o1} \cong Z_{o2}$ = one half of the rated balanced impedance.

V_{o1} and V_{o2} are the meter voltage readings corrected for voltage division

$$\% \text{ Unbalance} = \frac{2V_N}{\sqrt{2}\sqrt{V_{o1}^2 + V_{o2}^2 - 2V_N^2}}$$

Figure 1. Termination assembly.

4.5.2.3 Insertion VSWR. Testing shall be performed to demonstrate conformance to 3.2.2.3. Impedance at the input terminals of the properly terminated balun shall be measured by means of General Radio Company radio frequency bridge, type 916A, or equivalent. Each measured value shall be normalized and plotted on a Smith Chart, General Radio Co., form 756, or equal. The test shall be made at the frequencies specified in 4.5.2.1. The impedance plot is to fall within the circle of 1.25:1 VSWR. A sweep will also be performed to determine worse case frequency.

4.5.3 Weight and dimensions. Weight and dimensions of all units comprising the complete equipment shall be determined in accordance with MIL-E-16400 for compliance with 3.2.1.

4.5.5 Wind speed and icing. Conformance with the wind speed and icing requirements of 3.1.4 shall be demonstrated by engineering analysis. Calculations shall be submitted as part of the first article test report.

4.5.6 Enclosure test. The enclosure tests shall be conducted as specified in MIL-STD-108 for conformance to 3.1.5.5.

4.5.7 Temperature tests. Temperature tests shall be conducted in accordance with MIL-E-16400 for class 1 equipment in both nonoperating and operating conditions to demonstrate conformance to 3.1.4.

4.5.8 Humidity test. Equipment shall be subjected to the conditioning and test as specified in MIL-E-16400 for conformance to 3.1.4. The insertion VSWR test of 4.5.2.3 shall be performed at the following points throughout the humidity test:

(a) Following the conditioning specified in MIL-E-16400
(b) During the second 24-hour temperature cycle
(c) After 5 complete temperature cycles
(d) At the end of the 8-hour humidity cycle

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4.5.11 Shock, vibration and altitude. Tests shall be performed to demonstrate compliance with 3.1.5. The equipment shall be subjected to the shock test of MIL-STD-810, Method 516, Procedure II. The vibration test shall be in accordance with MIL-STD-810, Method 514, Procedure X, test level curve AB. With the equipment in the horizontal position, the altitude test shall be conducted in accordance with MIL-STD-810, Method 500, Procedure I, Steps 1, 2, 4, and 6. The equipment shall be in the nonoperating condition during test and shall be capable of normal operation immediately after the shock, vibration, and altitude tests are completed.

4.5.12 Salt spray test. The salt spray test shall be performed in accordance with MIL-E-16400 to demonstrate compliance with 3.1.8 and 3.1.10.

4.5.13 Thermal design. Calculations proving compliance with the thermal design requirement of MIL-E-16400 shall be submitted as part of the first article test results. The calculations shall logically present, step by step, the procedure used in the heat transfer design or analysis to arrive at a satisfactory design. The calculations, with narrations as necessary to make the presentation clear, shall include, but not be limited to; reasons for selection of particular components, their maximum temperature limitations, reasons for selection of particular coolants and methods of heat transfer, and an overall heat flow description including the individual mechanisms of heat flow from generation to dissipation. The origin of all formulas used shall be indicated.

4.5.14 Solar radiation test. The equipment shall be subjected to the sunshine test of MIL-STD-810, Method 505, Procedure II to demonstrate compliance with 3.1.4, except the test duration shall be one 24-hour cycle in lieu of five continuous 24-hour cycles. The insertion VSWR test of 4.5.2.3 shall be performed during the period of maximum solar radiation intensity.

4.5.15 Intermodulation products. Measurements to determine agreement with 3.2.2.8 shall be made at the following approximate frequencies. Variations in these frequencies will be permitted, if necessary to avoid spurious responses in the measuring equipment. Only one output circuit need be measured, providing it can be demonstrated that all outputs are identical. All measurements shall be made with the input filter in the circuit, unless otherwise noted.

f_1 (MHz)	f_2 (MHz)	$f_1 \pm f_2$ (MHz)	$2f_1 \pm f_2$ (MHz)
2.1	3.0	5.1 +	7.2 +
3.5	5.0	1.5 -	2.0 -
3.5	5.0	8.5 +	12.0 +
5.6	8.0	13.6 +	19.2 +
5.6	8.0	2.4 -	3.2 -
7.7	11.0	18.7 +	26.4 +
7.7	11.0	3.3 -	4.4 -
11.2	16.0	27.2 +	6.4 -
11.2	16.0	4.8 -	53.4 +
15.4	22.0	6.6 -	8.8 -
21.0	30.0	9.0 -	12.0 -
3.5	3.0	0.5 -	10.0 +
2.2	3.0	0.8 -	1.4 -
15.4	22.0	37.4 +	52.8 +
30.0	21.0	51.0 +	39 -
21.0	30.0	51.0 +	72 +
0.7	0.3	0.4 -	1.7 [1/] +
0.8	0.5	0.3 -	1.1 [1/] -

[1/] To be performed with input filter bypassed.

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4.6 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. Equipments purchased under this specification are intended for installation at Naval Communications Stations worldwide. The equipment will provide the interface between the unbalanced output of a receiver and the balanced input of an antenna, and will normally be mounted outdoors in a completely unsheltered environment.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements:

(a) Title, number and date of this specification
(b) Identification and markings, if other than specified (see 3.1.11)
(c) Quantity of first article sample units required (see 4.3)
(d) Marking, levels of preservation and packaging and packing required (see 5.1)

6.2.2 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to the following:

(a) Inspection system plan (see 4.1.1)
(b) Test procedures (see 4.1.3)
(c) First article test report (see 4.3.1)
(d) Quality conformance inspection report (see 4.4.6)

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Field change kits. If field change kits are necessary to correct latent or design defects subsequent to equipment delivery, such kits shall be in accordance with MIL-F-17655.

Preparing activity:
Navy - EC

(Project 5950-N108)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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